Sparse Autoencoder-Based Identification and Causal Ablation of Deceptive Circuits in Large Language Models

Abstract. Large language models can produce strategically misleading outputs—a safety failure known as deceptive alignment. Existing mechanistic interpretability tools identify residual-stream features but lack a method to pinpoint and intervene on the circuits that cause deception. We propose (1) training sparse autoencoders (SAEs) on residual-stream activations of GPT-2 XL and Pythia-6.9B, (2) ranking SAE features by pointwise mutual information with token-level deception labels, and (3) causally ablating the minimal deception-associated circuit via activation patching. Experiments on TruthfulQA will test whether targeted ablation reduces deceptive completions by $\geq$30% while increasing perplexity by <5%. Deliverables include an open-source pipeline, reproducibility package, and quantitative evidence on the causal footprint of deceptive circuits.

Keywords: mechanistic interpretability, sparse autoencoders, deceptive alignment, activation patching, circuit ablation

1. Introduction: Motivation, Gap, and Prior Work

Deceptive alignment—where a model produces false or misleading outputs it "knows" are incorrect—poses a concrete AI safety risk. Lin et al. [5] demonstrate with TruthfulQA that frontier models consistently fail truthfulness tests across 38 falsehood categories. Hubinger et al. [4] show theoretically that learned optimizers can acquire deceptive strategies during training. Yet no mechanistic method currently pinpoints which internal circuits execute deception or supports targeted intervention.

Sparse autoencoders have emerged as the tool of choice for decomposing polysemantic neurons: Cunningham et al. [2] show that SAE dictionaries trained on residual-stream activations recover monosemantic, human-interpretable features; Gao et al. [3] demonstrate that scaling SAEs yields high-fidelity feature coverage with minimal reconstruction error. On the circuit side, Conmy et al. [1] develop automated circuit discovery via activation patching and edge attribution patching, but focus on syntactic tasks (Indirect Object Identification) rather than behavioral safety properties. Marks et al. [6] identify linear "truth directions" in residual-stream geometry using linear probes, but do not use SAE decomposition or causal ablation.

Gap: SAE feature extraction and circuit-level causal ablation exist as separate techniques, but no work combines them to identify and suppress the minimal circuit causally responsible for deceptive outputs. This proposal closes that gap.

2. Project Goal and Hypotheses

Goal: Produce a reproducible pipeline that (i) trains SAEs on residual-stream activations, (ii) selects the minimal SAE features causally associated with deceptive token patterns via PMI ranking, and (iii) ablates the corresponding neurons to suppress deception at inference time.

- H1 Targeted ablation reduces deceptive completion frequency on TruthfulQA by $\geq$30% relative to the unmodified baseline while increasing perplexity by <5%.
- H2 The reduction is attributable to $\leq$5% of SAE features (those with highest PMI against deception labels), as confirmed by counterfactual ablation analysis.
- H3 The identified circuit and ablation protocol generalize across model scales (GPT-2 XL $\rightarrow$ Pythia-6.9B), yielding Pearson $r \geq 0.6$ between activated deception-associated feature count and measured deception rate across out-of-distribution prompt families.

3. Methods: Technical Approach and Agent Workflow

The pipeline (Figure 1) proceeds through five stages with defined inputs, outputs, and stopping criteria.

Stage 1 — Data and Activation Collection. Input: TruthfulQA (817 questions). Process: TransformerLens hooks capture layer-wise residual-stream activations for each prompt; generated answers are labeled deceptive/honest using the TruthfulQA grading rubric. Output: Activation tensor $\mathbf{A} \in \mathbb{R}^{N \times L \times d}$ and binary deception labels $y \in \{0, 1\}^N$.

Stage 2 — SAE Training. Input: $\mathbf{A}$. Process: SAELens trains a sparse autoencoder with bottleneck at 2% of hidden dimension d, sparsity coefficient 0.1, for 50 k steps (following [2]). Output: Dictionary $\mathbf{D} \in \mathbb{R}^{k \times d}$ of k learned features and per-token sparse activation maps $\mathbf{Z} \in \mathbb{R}^{N \times k}$.

Stage 3 — Feature Selection. Input: $\mathbf{Z}$, labels y. Process: Pointwise mutual information $\mathrm{PMI}(f_i; y)$ is computed for each feature f_i; features are ranked and the smallest subset covering 80% of total PMI mass is retained. Output: Deception-associated feature set $\mathcal{F}^*$ (expected $|\mathcal{F}^*| \leq 0.05k$).

Stage 4 — Causal Ablation. Input: $\mathcal{F}^*$, held-out test split (317 questions). Process: Activation patching replaces each selected feed-forward neuron's activation with its mean over non-deceptive examples. The patched model generates answers on the held-out set. Output: Deceptive completion rate δ and perplexity π on held-out set. Stopping criterion: $\delta_{\text{ablated}}/\delta_{\text{baseline}} \leq 0.7$ (i.e., $\geq$30% reduction) and $\pi_{\text{ablated}}/\pi_{\text{baseline}} < 1.05$.

Stage 5 — Generalization. Input: Three OOD prompt families (political misinformation, medical myth, financial fraud; $\approx$200 prompts each). Process: Activated $\mathcal{F}^*$ feature count is correlated with observed deception rate; Pearson r computed. Output: Generalization coefficient r for H3.

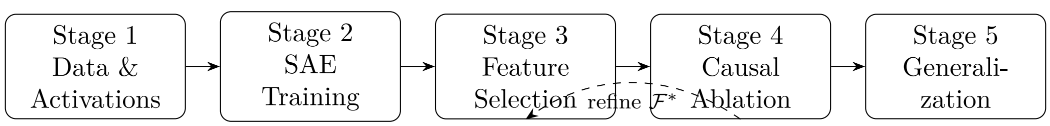

Figure 1: Five-stage pipeline. Dashed arrow: if $\geq$30% deception reduction is not achieved after Stage 4, Feature Selection threshold is relaxed and ablation is retried (feedback loop).

4. Expected Results and Research Milestones

- Weeks 1–4: SAE trained on GPT-2 XL; feature-selection pipeline implemented; deception-feature vocabulary characterized. Artifact: feature activation maps + PMI ranking.
- Weeks 5–8: Causal ablation on TruthfulQA held-out set; H1 and H2 evaluated. Artifact: ablation results table with deception rates and perplexity deltas.
- Weeks 9–12: Cross-scale validation (Pythia-6.9B); OOD generalization; H3 evaluated. Final paper and open-source release. Artifact: reproducibility package (code, seeds, checkpoints).

Expected outcomes: a circuit-level account of which SAE features causally mediate deception, quantitative evidence that targeted ablation suppresses deception without broad capability degradation, and a general-purpose pipeline applicable to other behavioral safety targets.

5. Evaluation Plan

Primary safety metric: deceptive completion rate on TruthfulQA held-out split (317 questions), compared to unmodified baseline and a random-ablation control (random $|\mathcal{F}^*|$ features ablated). Secondary metric: perplexity on a held-out WikiText-103 slice to measure language-modeling quality. Statistical tests: two-sided Wilcoxon signed-rank test ($\alpha = 0.05$) for deceptive rate; paired t-test for logit differences on deceptive tokens. Results reported with 95% confidence intervals. Mechanistic validation: counterfactual ablation of each $f_i \in \mathcal{F}^*$ in isolation; feature must produce statistically significant logit drop for deceptive tokens to remain in $\mathcal{F}^*$. Generalization: Pearson r between activated feature count and deception rate across three OOD prompt families ($\geq$60 examples each).

6. Risks and Mitigation

- GPU memory: SAE training and activation patching on 6.9B-parameter models may exceed 24 GB VRAM. Mitigation: prototype on GPT-2 XL first; apply 8-bit quantization and gradient checkpointing before scaling; allocate overflow A100 burst capacity.
- Dataset scope: TruthfulQA covers factual falsehoods, not strategic deception. Mitigation: three hand-crafted OOD prompt families; deception operationalized as false output where linear probe [6] confirms correct internal representation, ensuring behavioral precision.
- Over-ablation: removing too many features could degrade general language modeling. Mitigation: enforce perplexity cap (<5% increase); roll back to the last feature subset that satisfies the cap; report Pareto curve of deception reduction vs. perplexity cost.

7. Resources, Tools, Budget, and Release Plan

Compute: 2× NVIDIA A100 (40 GB) for SAE training (≈6 GPU-hours on GPT-2 XL); cloud burst credits for Pythia-6.9B runs (≈18 GPU-hours). Software: TransformerLens (circuit analysis), SAE-Lens (SAE training), PyTorch. Data: TruthfulQA (open-access); OOD prompts hand-constructed (≈600 total). Release plan: all code, hyperparameters, random seeds, pre-trained SAE checkpoints, and evaluation scripts released under MIT license upon acceptance; feature activation maps released as a Hugging Face dataset.

References

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